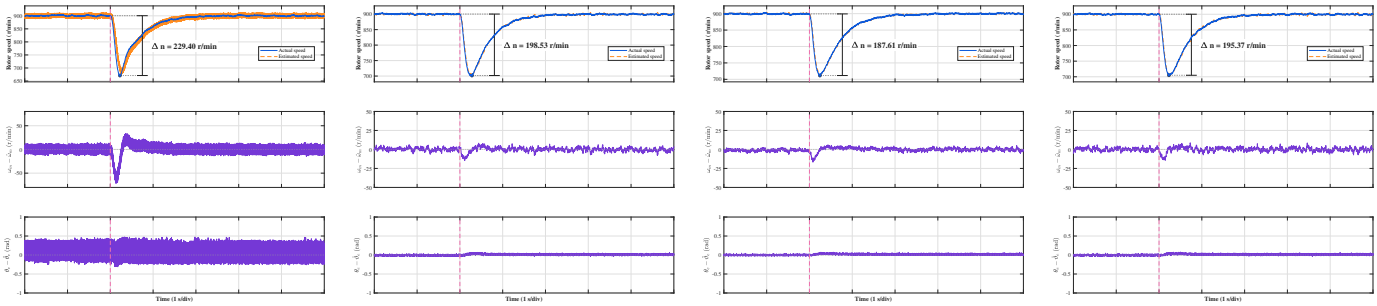
Supplementary Material for Cascaded Prescribed-Time Super-Twisting Observer for PMSM Sensorless Drive Systems

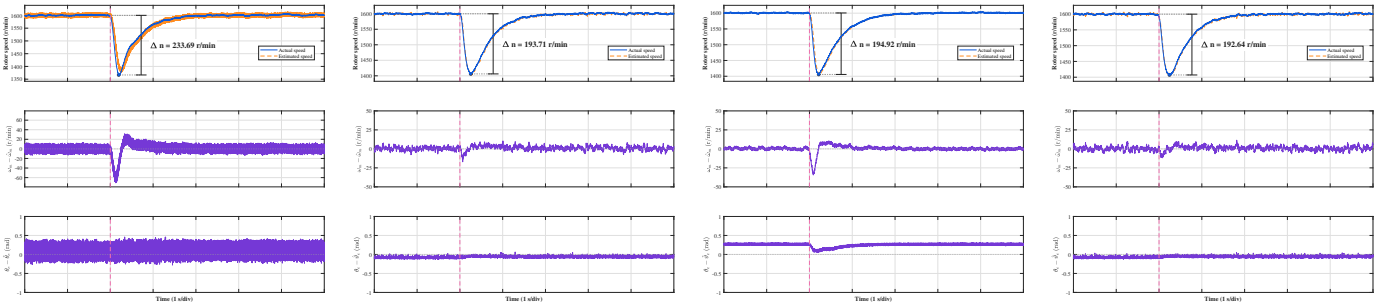
Load-Disturbance Tests

A load torque of 0.1 N·m is applied after the drive reaches sensorless closed-loop steady state. Figs. S1 and S2 present the responses at 900 and 1600 r/min, respectively. Method names are consistent with the main paper.



(a) SMO (b) FT-STO (c) FxT-STO (d) PT-STO

Fig. S1. Experimental responses to a 0.1 N·m load step applied after steady-state operation at 900 r/min.

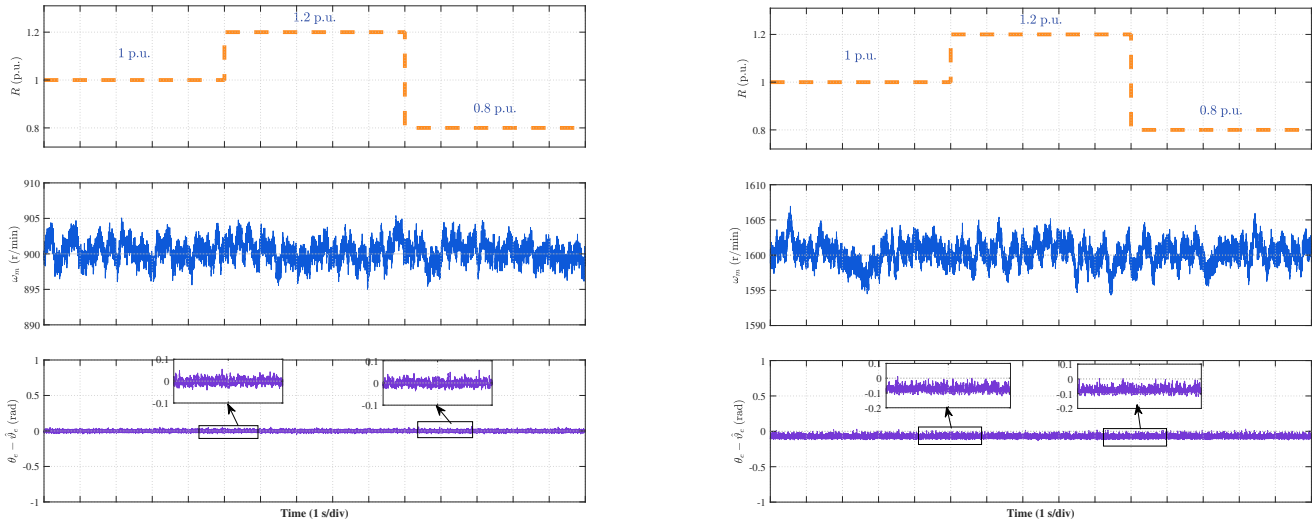


(a) SMO (b) FT-STO (c) FxT-STO (d) PT-STO

Fig. S2. Experimental responses to a 0.1 N·m load step applied after steady-state operation at 1600 r/min.

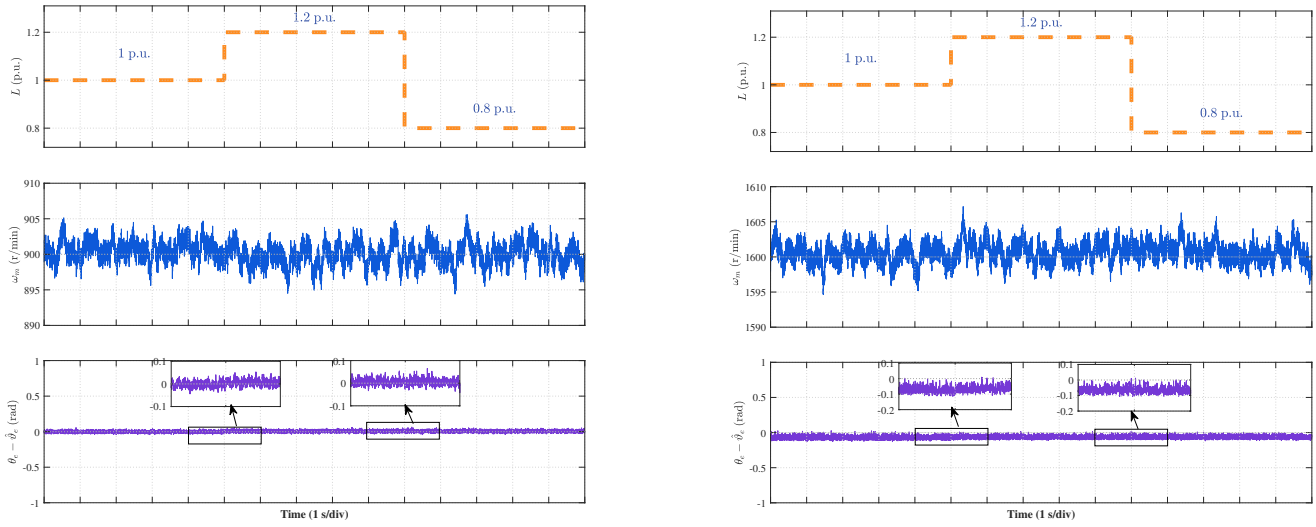
OBSERVER-PARAMETER MISMATCH TESTS

The resistance or inductance used in the PT-STO is varied individually from 1.0 to 1.2 and then to 0.8 times its nominal value, while the other parameter remains nominal. Each stage lasts 5 s. Figs. S3 and S4 show the corresponding responses.



(a) 900 r/min (b) 1600 r/min

Fig. S3. Experimental responses to observer resistance mismatch at 900 and 1600 r/min.



(a) 900 r/min (b) 1600 r/min

Fig. S4. Experimental responses to observer inductance mismatch at 900 and 1600 r/min.